

3D Scatter Plot Projection Report (Ultra-Concise)

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1. Project Overview

This project visualizes 3D data points in a web browser, allowing users to interact with rotations and different projection styles.

2. Workflow of the Project

It starts with generating random 3D points, applying rotations, converting to 2D screen coordinates, and then drawing them on an HTML5 Canvas.

3. Random Number Generation for 3D Points

Random X, Y, Z coordinates are created within a set range to populate the 3D space dynamically for visualization.

4. Rotation Mathematics

Specific mathematical formulas are used to rotate points around the X (Pitch), Y (Yaw), and Z (Roll) axes.

4.1 Yaw Rotation

This rotates points around the Y-axis, changing X and Z coordinates to simulate horizontal turning.

4.2 Pitch Rotation

Points rotate around the X-axis, affecting Y and Z coordinates for an up-and-down tilting effect.

4.3 Roll Rotation

Rotation around the Z-axis alters X and Y coordinates, creating a side-to-side rolling motion.

5. Degree-to-Radian Conversion

Angles given in degrees are converted to radians for compatibility with JavaScript's trigonometric functions.

6. Orthographic Projection

This projection maps 3D points to 2D by simply dropping the Z-coordinate, maintaining object sizes without depth perception.

7. Perspective Projection

Simulating human vision, this projection makes distant objects appear smaller, adding realism and depth to the 2D display.

8. Implementation Details

The project uses HTML for structure, CSS for styling, and JavaScript with HTML5 Canvas for all the logic and rendering.

9. Complete Working of the Project

From page load, points are generated, transformed, and continuously redrawn on the canvas, responding to user controls for an animated view.

10. Conclusion

This project demonstrates 3D visualization principles using web technologies, offering insights into rotations, projections, and interactive graphics.